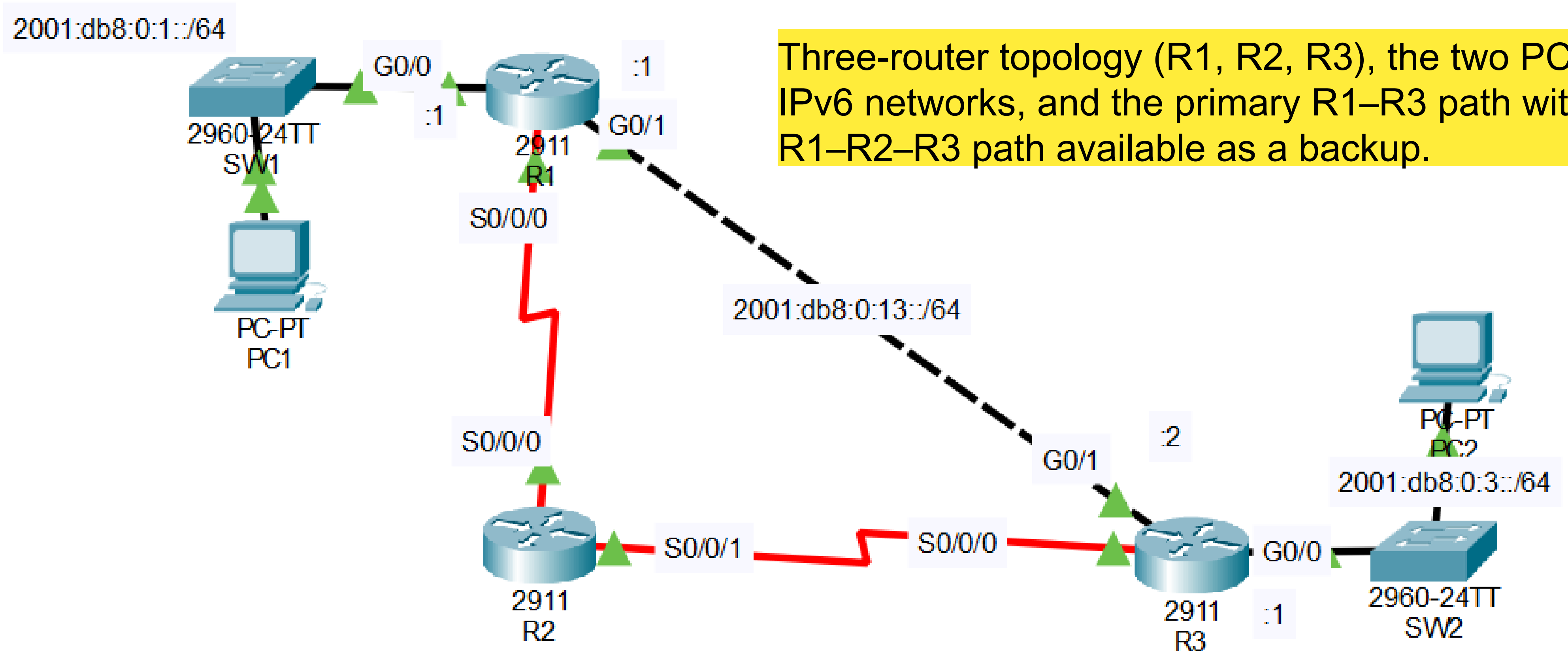

Three-router topology (R1, R2, R3), the two PC LANs, IPv6 networks, and the primary R1–R3 path with the R1–R2–R3 path available as a backup.

IPv6 addresses have been pre-configured on the routers.
The serial connections use link-local addresses only.

1. Enable IPv6 routing on each router.

2. Use SLAAC to configure IPv6 addresses on the PCs.
   What IPv6 address was configured on each PC?

3. Configure static routes on the routers to allow PC1 and PC2 to ping each other.
   The path via R2 should be used only as a backup path.

R1

IOS Command Line Interface

Enabled IPv6 routing on R1.

```
export@cisco.com.

Cisco CISCO2911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
3 Gigabit Ethernet interfaces
2 Low-speed serial(sync/async) network interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!


%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state
to up

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state
to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up


R1>
R1>en
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ipv6 unicast-routing
R1(config)#
```

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Top

R2

Physical Config CLI Attributes

IOS Command Line Interface

```
If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISCO2911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
3 Gigabit Ethernet interfaces
2 Low-speed serial(sync/async) network interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!


%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

%LINK-5-CHANGED: Interface Serial0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1, changed state to up


R2>
R2>en
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#ipv6 unicast-routing
R2(config)#
```

Enabled IPv6 routing on R2.

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R3

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
Cisco CISCO2911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
3 Gigabit Ethernet interfaces
2 Low-speed serial(sync/async) network interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!


%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state
to up

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state
to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up


R3>
R3>
R3>en
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#ipv6 unicast-routing
R3(config)#
```

Enabled IPv6 routing on R3.

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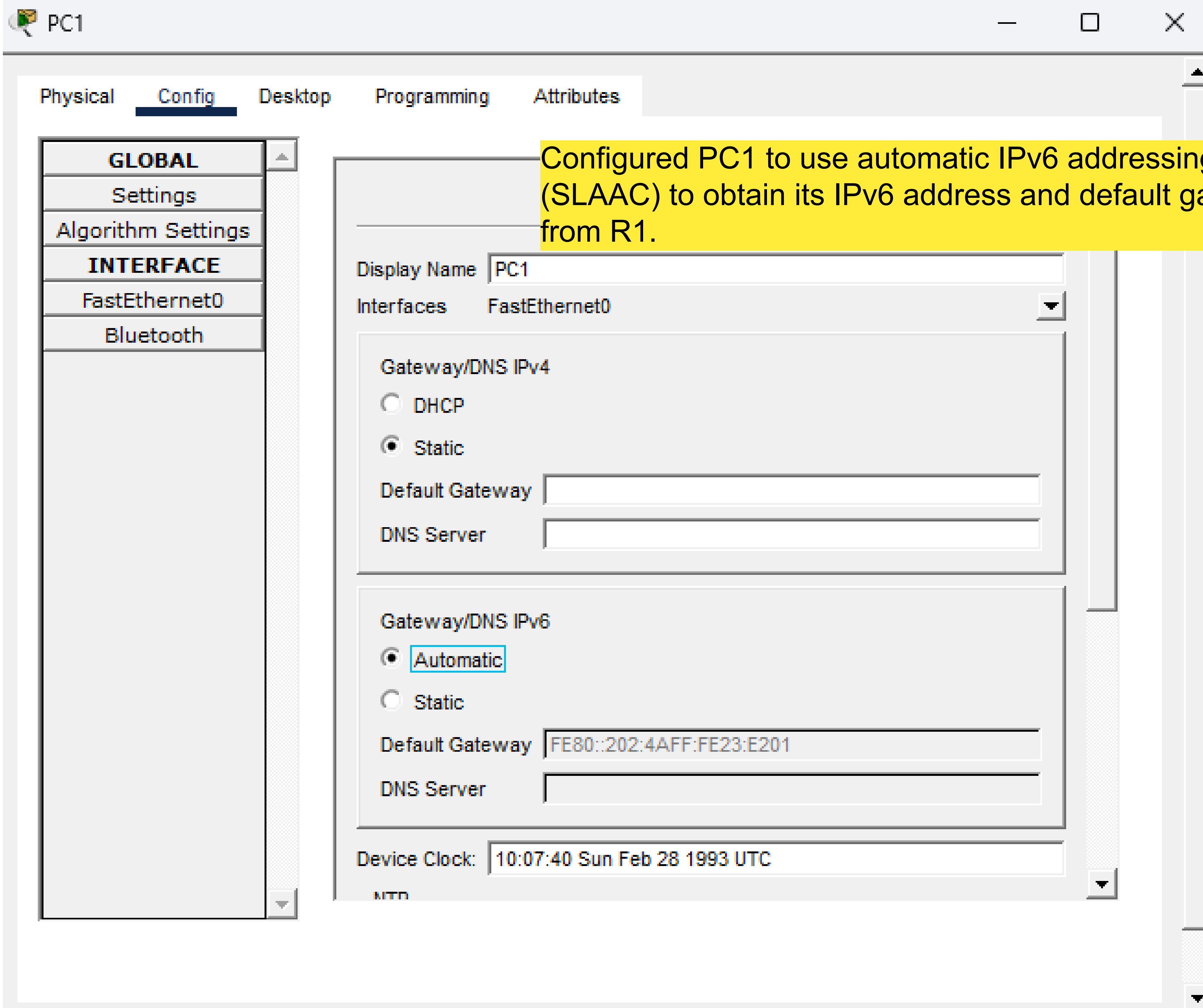
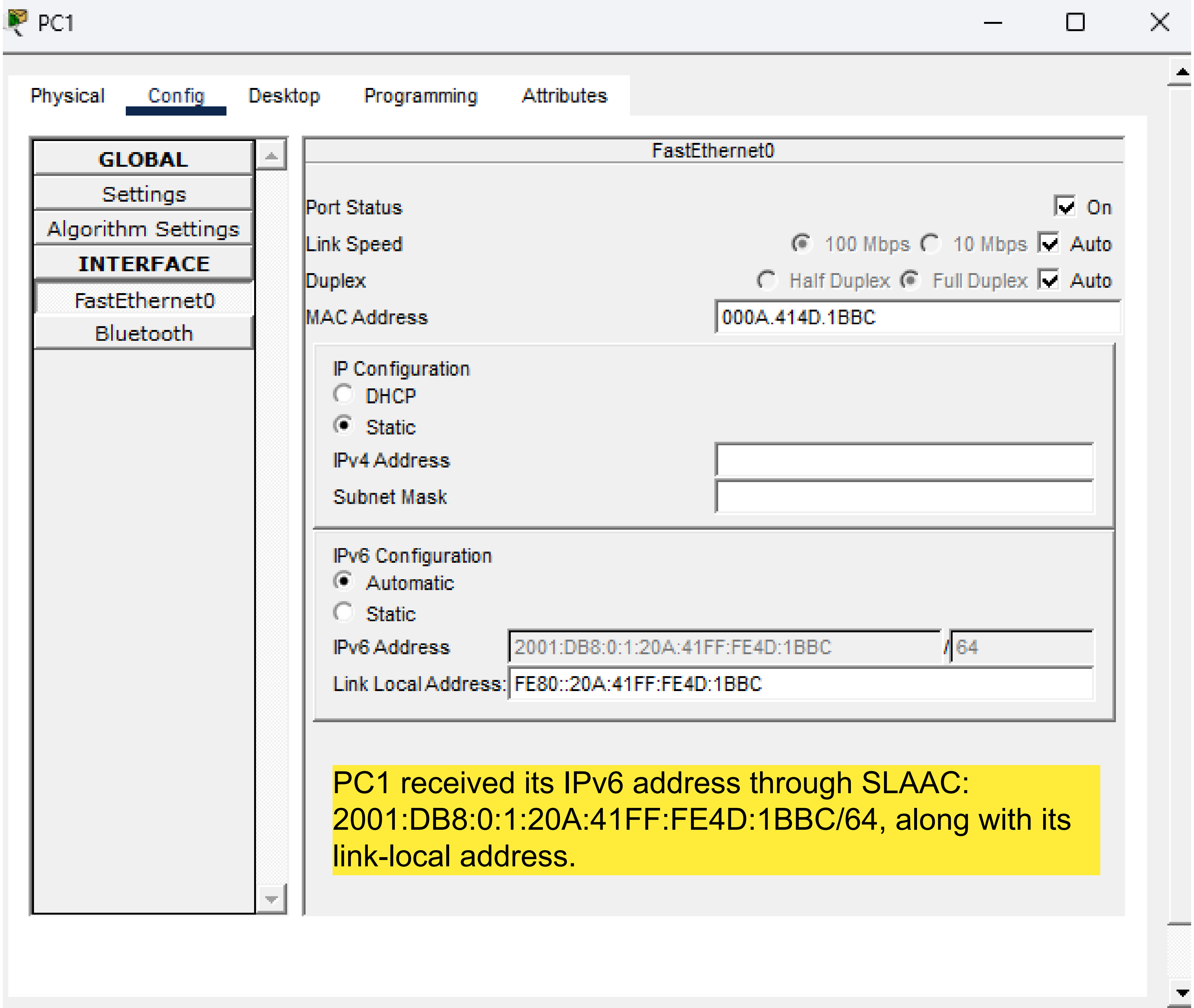
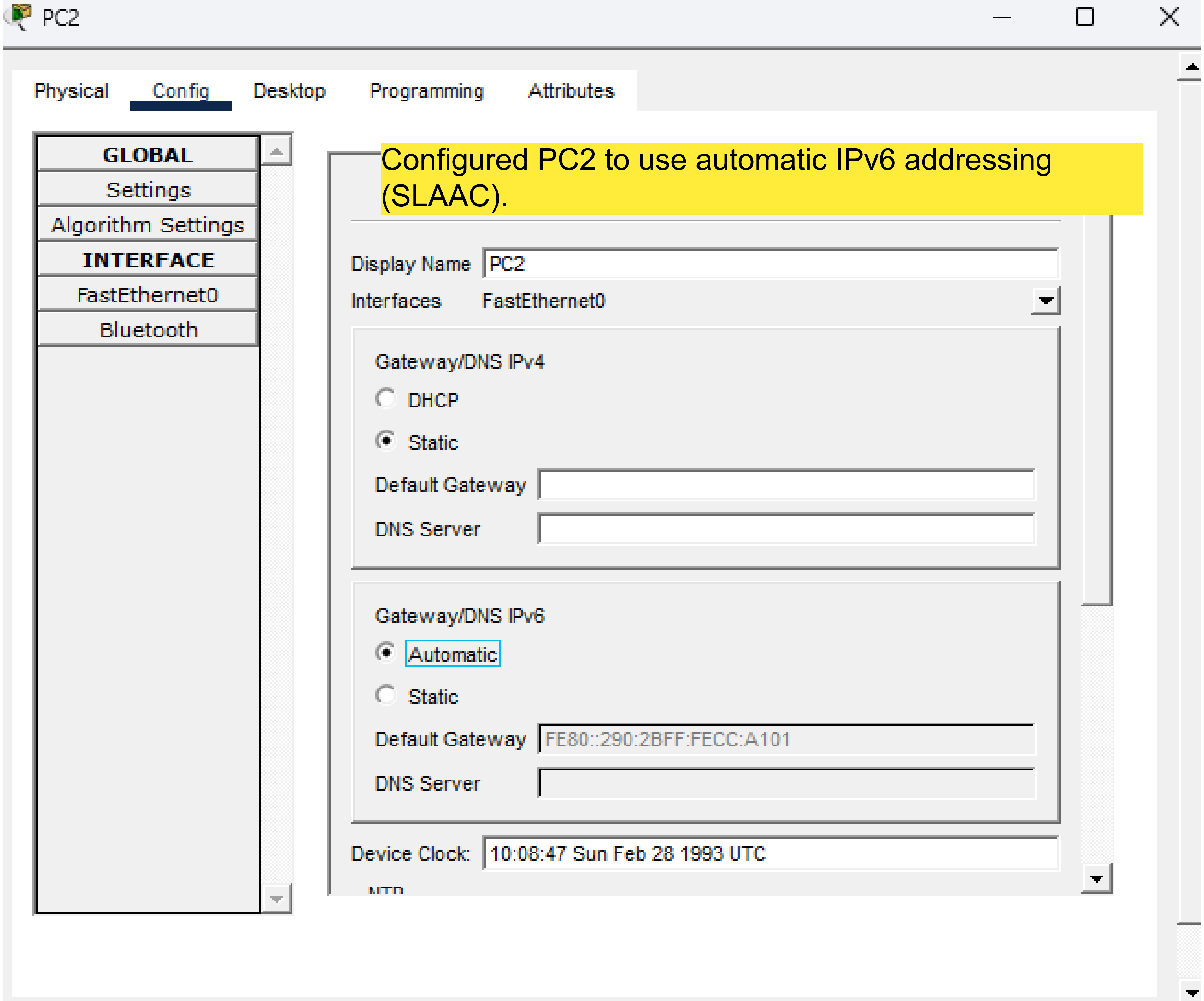
Configured PC1 to use automatic IPv6 addressing (SLAAC) to obtain its IPv6 address and default gateway from R1.

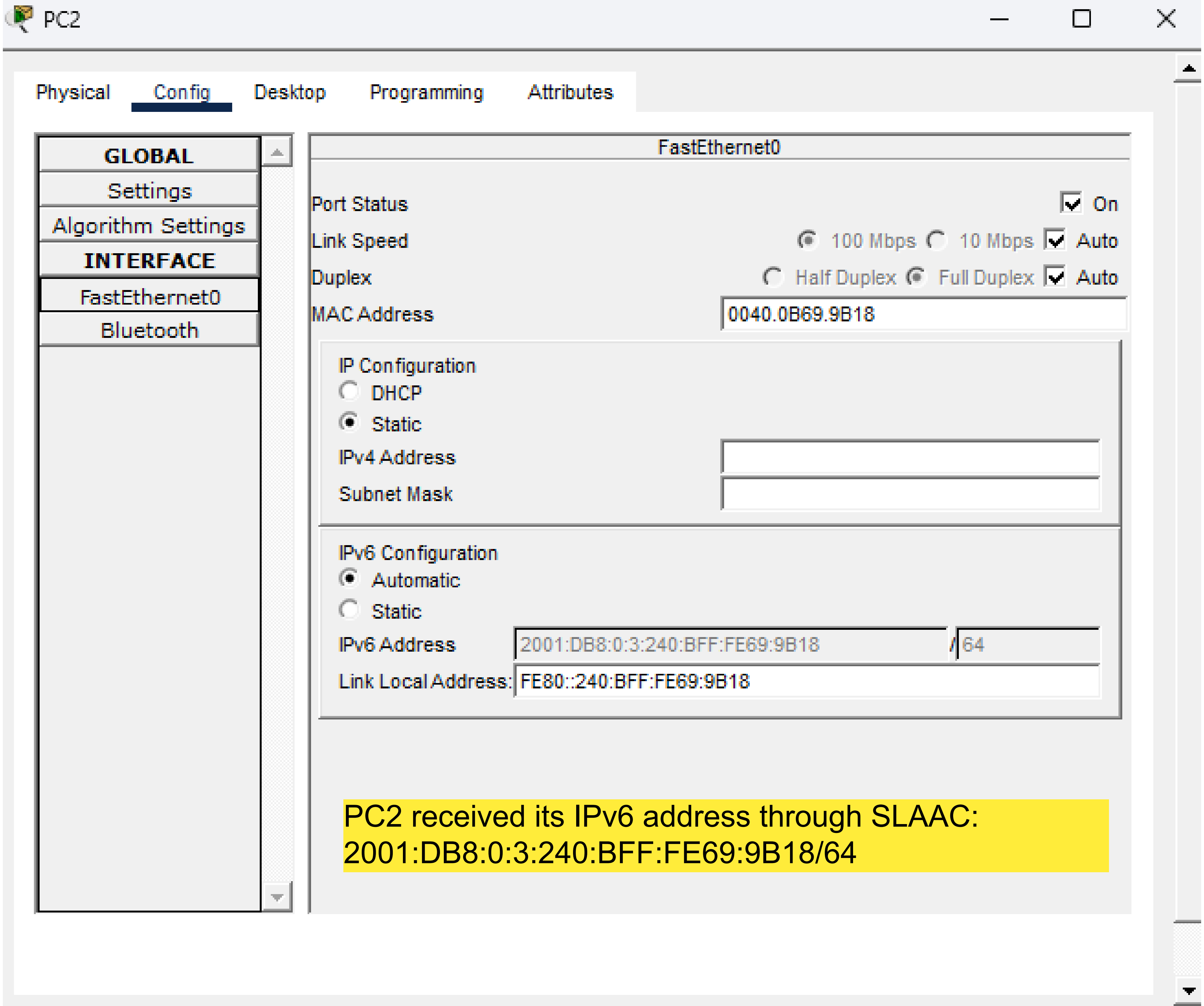




PC1 received its IPv6 address through SLAAC: 2001:DB8:0:1:20A:41FF:FE4D:1BBC/64, along with its link-local address.



Configured PC2 to use automatic IPv6 addressing (SLAAC).


PC2
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
FastEthernet0
Port Status
On
Link Speed
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0040.0B69.9B18
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
IPv6 Configuration
Automatic
Static
IPv6 Address
2001:DB8:0:3:240:BFF:FE69:9B18
/
64
Link Local Address:
FE80::240:BFF:FE69:9B18
PC2 received its IPv6 address through SLAAC:
2001:DB8:0:3:240:BFF:FE69:9B18/64

Configured a static IPv6 route on R1 to reach PC2's network through R3.

```
R1>
R1>en
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ipv6 unicast-routing
R1(config)#
R1(config)#
R1(config)#
R1(config)#ipv6 route 2001:db8:0:3::/64 g0/1 2001:db8:0:13::2
R1(config)#
```

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R2

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINK-5-CHANGED: Interface Serial0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1, changed state to up


R2>
R2>en
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#ipv6 unicast-routing
R2(config)#
R2(config)#
R2(config)#
R2(config)#do sh ipv6 int br
GigabitEthernet0/0          [administratively down/down]
    unassigned
GigabitEthernet0/1          [administratively down/down]
    unassigned
GigabitEthernet0/2          [administratively down/down]
    unassigned
Serial0/0/0                 [up/up]
    FE80::20B:BEFF:FED7:4901
Serial0/0/1                 [up/up]
    FE80::20B:BEFF:FED7:4901
Vlan1                       [administratively down/down]
    unassigned
R2(config)#
```

Checked R2's link-local IPv6 address to use as the next hop for R1's backup route.

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R1

Physical Config CLI Attributes

IOS Command Line Interface

```
R1(config)#ipv6 unicast-routing
R1(config)#
R1(config)#
R1(config)#
R1(config)#ipv6 route 2001:db8:0:3::/64 g0/1 2001:db8:0:13::2
R1(config)#
R1(config)#ipv6 route 2001:db8:0:3::/64 s0/0/0 FE80::20B:BEFF:FED7:4901
R1(config)#ipv6 route 2001:db8:0:3::/64 s0/0/0 FE80::20B:BEFF:FED7:4901 5
R1(config)#do sh ipv6 route
IPv6 Routing Table - 6 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
C   2001:DB8:0:1::/64 [0/0]
     via GigabitEthernet0/0, directly connected
L   2001:DB8:0:1::1/128 [0/0]
     via GigabitEthernet0/0, receive
S   2001:DB8:0:3::/64 [1/0]
     via 2001:DB8:0:13::2, GigabitEthernet0/1
C   2001:DB8:0:13::/64 [0/0]
     via GigabitEthernet0/1, directly connected
L   2001:DB8:0:13::1/128 [0/0]
     via GigabitEthernet0/1, receive
L   FF00::/8 [0/0]
     via Null0, receive
R1(config)#
```

Configured a backup static route on R1 through R2 with an administrative distance of 5.

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R3

IOS Command Line Interface

Checked R3's link-local IPv6 address for use as the next hop on R2.

```
R3>
R3>en
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#do sh ipv6 int br
GigabitEthernet0/0         [up/up]
    FE80::290:2BFF:FECC:A101
    2001:DB8:0:3::1
GigabitEthernet0/1         [up/up]
    FE80::290:2BFF:FECC:A102
    2001:DB8:0:13::2
GigabitEthernet0/2         [administratively down/down]
    unassigned
Serial0/0/0                [up/up]
    FE80::290:2BFF:FECC:A101
Serial0/0/1                [administratively down/down]
    unassigned
Vlan1                      [administratively down/down]
    unassigned
R3(config)#
```

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Top

R2

Physical Config CLI Attributes

IOS Command Line Interface

```
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#ipv6 route 2001:db8:0:13::/64 s0/0/0 FE80::202:4AFF:FE23:E201
R2(config)#ipv6 route 2001:db8:0:13::/64 s0/0/0 FE80::290:2BFF:FECC:A101^Z
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#en
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#ipv6 route 2001:db8:0:1::/64 s0/0/0 FE80::202:4AFF:FE23:E201
R2(config)#ipv6 route 2001:db8:0:3::/64 s0/0/1 FE80::290:2BFF:FECC:A101
R2(config)#do sh ipv6 route
IPv6 Routing Table - 4 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
S   2001:DB8:0:1::/64 [1/0]
     via FE80::202:4AFF:FE23:E201, Serial0/0/0
S   2001:DB8:0:3::/64 [1/0]
     via FE80::290:2BFF:FECC:A101, Serial0/0/1
S   2001:DB8:0:13::/64 [1/0]
     via FE80::202:4AFF:FE23:E201, Serial0/0/0
L   FF00::/8 [0/0]
     via Null0, receive
R2(config)#
```

Configured static routes on R2 to reach both PC1 and PC2 networks.

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R3

IOS Command Line Interface

Configured the primary and backup static routes on R3 to reach PC1’s network.

```
R3>
R3>en
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#do sh ipv6 int br
GigabitEthernet0/0         [up/up]
    FE80::290:2BFF:FECC:A101
    2001:DB8:0:3::1
GigabitEthernet0/1         [up/up]
    FE80::290:2BFF:FECC:A102
    2001:DB8:0:13::2
GigabitEthernet0/2         [administratively down/down]
    unassigned
Serial0/0/0                [up/up]
    FE80::290:2BFF:FECC:A101
Serial0/0/1                [administratively down/down]
    unassigned
Vlan1                      [administratively down/down]
    unassigned
R3(config)#
R3(config)#
R3(config)#
R3(config)#
R3(config)#ipv6 route 2001:db8:0:1::/64 g0/1 2001:db8:0:13::1
R3(config)#ipv6 route 2001:db8:0:1::/64 s0/0/0 FE80::20B:BEFF:FED7:4901 5
R3(config)#
```

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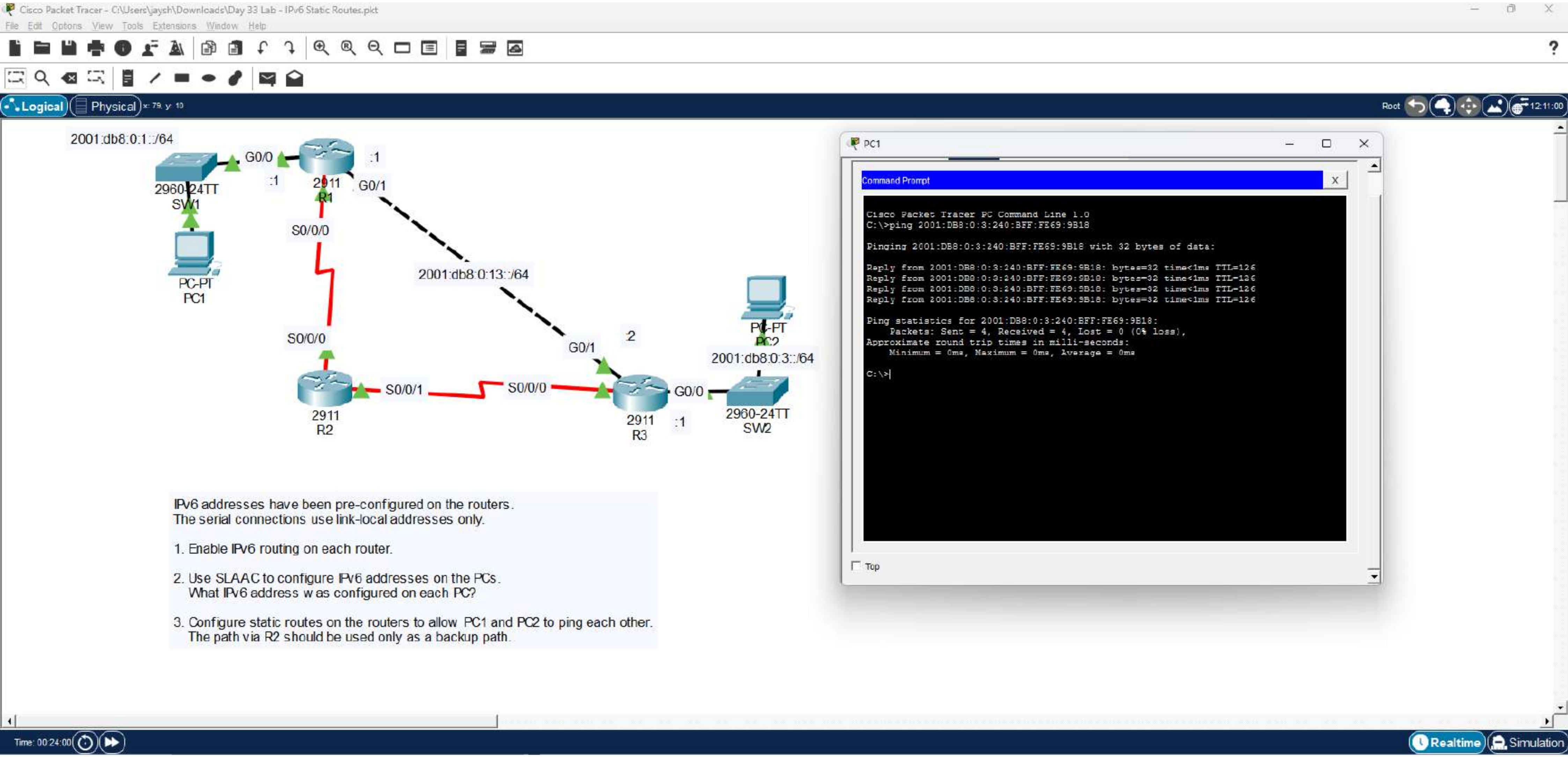
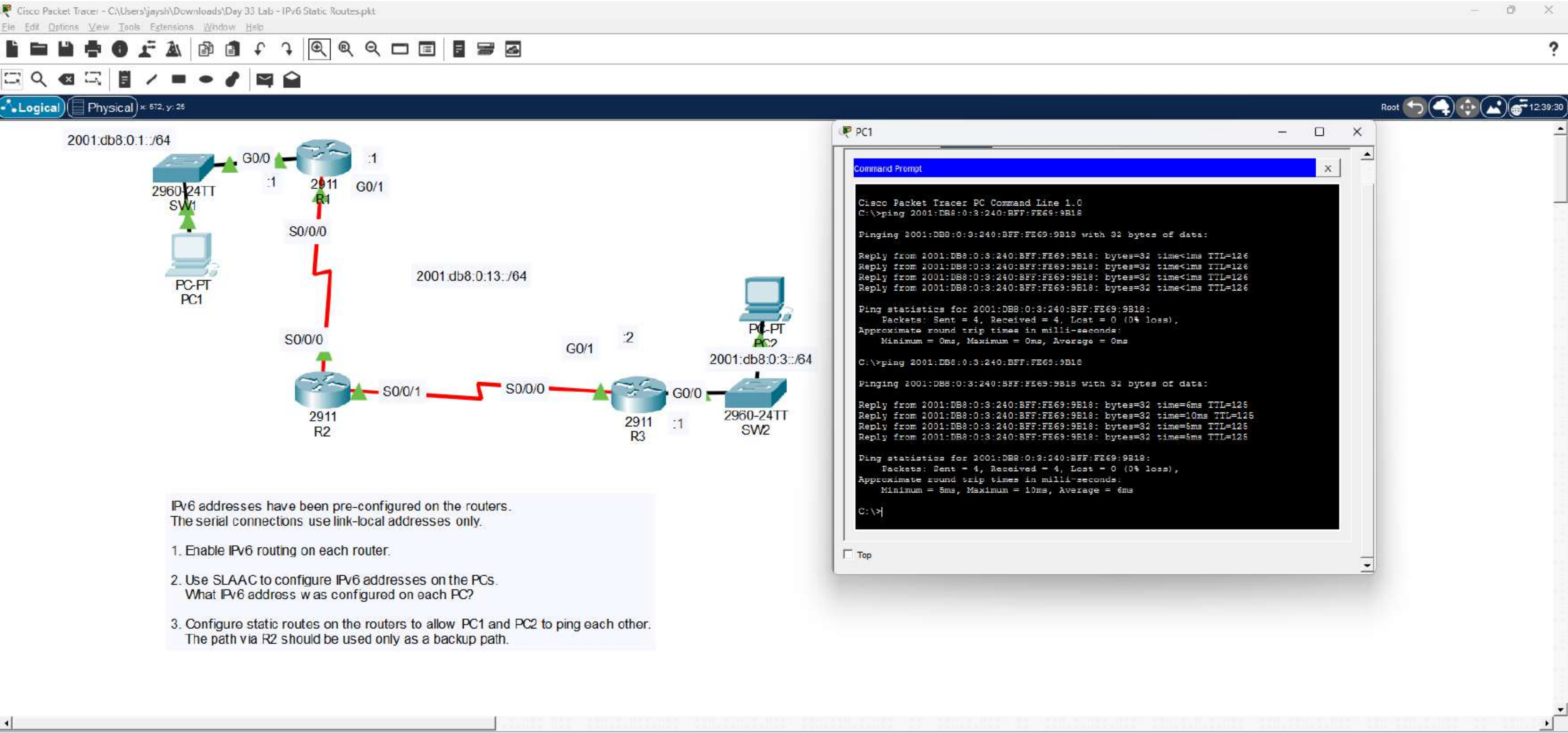
Top

Successfully pinged PC2 from PC1 through the configured IPv6 static route.



Tested the backup route by removing the primary R1–R3 connection and successfully pinging PC2 through R2.



Used tracert to verify the backup path through R2 after the primary R1–R3 connection was removed.

```
C:\>
C:\>tracert 2001:DB8:0:3:240:BFF:FE69:9B18

Tracing route to 2001:DB8:0:3:240:BFF:FE69:9B18 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      2001:DB8:0:1::1
  2   *         *         *         Request timed out.
  3   5 ms      0 ms      4 ms      2001:DB8:0:3::1
  4   4 ms      2 ms      2 ms      2001:DB8:0:3:240:BFF:FE69:9B18

Trace complete.

C:\>
```